Prodigy Anesthesia Quick Review

Subarachnoid Block

How many pairs of spinal nerves are there and how are they grouped anatomically?	There are 31 pairs of spinal nerves: 8 cervical, 12 thoracic, 5 lumbar, 5 sacral, and 1 coccygeal. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 9.*
What is the only absolute contraindications to spinal anesthesia?	Patient refusal *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 929.*
From exterior to interior, name the structures the spinal needle passes through when performing a subarachnoid block via the midline approach.	Skin, subcutaneous tissue, supraspinous ligament, interspinous ligament, ligamentum flavum, dura mater, arachnoid mater, and subarachnoid space. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 791.*
From exterior to interior, what structures will the needle traverse when performing a subarachnoid block using a paramedian approach?	Skin, subcutaneous tissue, paraspinous muscle, ligamentum flavum, dura mater, arachnoid mater, and subarachnoid space. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 792.*
Where is cerebrospinal fluid created and absorbed?	CSF is created in the choroid plexus and absorbed in the arachnoid granulations (remember C for Create and Choroid and A for Absorb and Arachnoid) *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
What are some of the relative contraindications to spinal anesthetic?	Hypovolemia, sepsis, increased intracranial pressure, coagulopathy or thrombocytopenia, and infection at the puncture site. Other major contraindications include: conditions resulting in left ventricular outflow obstruction such as aortic stenosis or hypertrophic subaortic stenosis restrict the ability of the heart to increase cardiac output as compensation for hypotension due to the sympathectomy induced by neuraxial anesthesia. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 929.*
What are the major potential complications arising from a spinal anesthetic?	Neurologic dysfunction, allergic reaction, anterior spinal artery syndrome, trauma, drug toxicity, infection, hematoma, and total spinal blockade. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 166.*

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Subarachnoid Block

How far does the subarachnoid space extend caudally in adults?	The subarachnoid space extends from the foramen magnum to S2 in adults. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 943.*
How far does the subarachnoid space extend caudally in children?	The subarachnoid space extends from the foramen magnum to S3 in children. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 943.*
How far does the spinal cord extend caudally in children?	The spinal cord extends to L3 in children. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 943.*
How far does the spinal cord extend caudally in adults?	The spinal cord extends to L2 in adults. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1071.*
Where are the most dependent portions of the spinal column in the supine position?	Normally, in the supine position, the most dependent portion of the spinal column occurs at T4-T8 with the peak of the thoracolumbar curvature occurring at T4. This is important in relation to hyperbaric spinal solutions which will tend to pool in the T4-T8 curvature but limit their ascent above T4 in the supine position. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 956-957.*
Where are the two enlargements in the spinal cord and what causes them?	The cervical enlargement occurs at C4-T1 and the lumbar enlargement which occurs from L2-S3. The cervical enlargement is due to the nerve roots that produce the brachial plexus and the lumbar enlargement produces the lumbar plexus. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1072.*
What external anatomic landmark can be used to identify the L4 interspace?	The superior aspect of the iliac crest can be palpated at the level of the L4 interspace. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 1.*
What spinal landmark corresponds with the level of the posterior superior iliac spine?	S2 *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 1.*

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Subarachnoid Block

In what positions can a subarachnoid block be performed?	A subarachnoid block can be performed in the sitting, lateral, or prone positions. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1079.*
For what procedures is the sitting position for subarachnoid block advantageous?	The sitting position is advantageous for vaginal or urologic procedures where a 'saddle' block is preferable. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1079.*
For what procedures would it be advantageous to perform a subarachnoid block in the prone position?	The prone position is advantageous for rectal procedures because the patient can be placed into position for the surgery before the block is performed. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1079.*
For what procedures is it advantageous to perform a subarachnoid block in the lateral position?	The lateral position is advantageous for surgeries where you would prefer that the drug be concentrated on one side more than the other, such as hip surgery. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1079.*
How should the patient be instructed to change their posture to make performing a spinal anesthetic easier?	They should be encouraged to arch their back in the shape of a C which opens up the intervertebral spaces and makes insertion of the needle easier. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1079.*
Is it appropriate to wipe off excess Betadine with an alcohol prep prior to inserting the spinal needle for a subarachnoid block? Why or why not?	No, you should use a dry sterile gauze to wipe away excess Betadine after prepping the skin. Alcohol neutralizes the iodine solution and reduces its ability to function as an antiseptic. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1080.*
If Betadine is used as the antiseptic to clean the skin prior to performing a spinal anesthetic, how long should you wait after applying the Betadine before beginning the procedure and why?	You should wait at least 1 minute before beginning the spinal so that the Betadine has ample opportunity to dry. Then, you should wipe away any remaining liquid Betadine off of the skin. If Betadine (or any povidone-iodine solution) is introduced into the subarachnoid space, it can produce a chemical arachnoiditis. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1080.*
You are inserting a spinal needle during a subarachnoid block and feel a 'pop'. What might this indicate and what should you do next?	The 'pop' could represent the loss of resistance sensation felt as the needle penetrates beyond the ligamentum flavum and through dura and enters the subarachnoid space. You should withdraw the stylet and see if CSF flow is present. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 791.*

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During a subarachnoid block, you inserted the needle, felt a 'pop', removed the stylet and see CSF flow, what is the recommended step to take next?	It is recommended to rotate the needle 360 degrees in 90 degree increments to make sure that flow is constant in all four quadrants to ensure proper needle placement. Once you have verified the needle position, you can verify CSF flow by aspirating with the spinal syringe and begin administering the medication for the spinal anesthetic. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 791-792.*
How is a paramedian approach to a subarachnoid block performed?	A skin wheal is raised 1 cm lateral to the superior tip of the spinous process at the lumbar level you wish to access. The tissue below the skin wheal is infiltrated with local anesthetic. The introducer and needle are then inserted at a 10-15 degree angle toward midline. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 791-792.*
When performing spinal anesthesia, two 'pops' are often felt. The first pop encountered is due to the needle penetrating what structure?	The first 'pop' is the needle penetrating the ligamentum flavum and the second is the needle penetrating the dura mater. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 955.*
While inserting the spinal needle during a subarachnoid block, you feel as if you have struck bone. What should you do next?	Carefully check to see if the needle is midline. If the needle is directed away from midline, withdraw it and reposition the needle. If it appears to be in the midline, then 'walk' the needle caudally until the subarachnoid space is encountered. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 791.*
What are the most important factors that determine the spread of local anesthetics within the cerebrospinal fluid during a subarachnoid block?	The total dose of anesthetic, the site of injection, the baricity of the local anesthetic, and the position of the patient during and immediately after injection. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
What two factors determine the duration of action of a spinal anesthetic?	The anesthetic drug used and the total dose given. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
How does the volume of the local anesthetic injected affect the duration and spinal level achieved with the block?	Volumes between the ranges of 1 mL and 14 mL has been thoroughly tested and has little effect on the duration of action of the spinal anesthetic or the sensory level achieved. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1082.*

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How does the addition of a vasoconstrictor to a local anesthetic affect the duration of action of a spinal anesthetic?	The addition of a vasoconstrictor to the local anesthetic used in a subarachnoid block will prolong the duration of action of the block. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
What is the normal specific gravity of CSF?	1.004 to 1.009 *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
What is the difference between specific gravity and baricity?	Specific gravity is the comparison between the density of a solution (such as CSF) when compared to water, which is assigned a value of 1. A solution that has a higher density than water has a value higher than 1. A solution with a density lower than water has a specific gravity less than 1. Baricity is the relationship of one solution's density to the density of another solution. For example if one solution is more dense than another, it is said to be hyperbaric in relation to that solution. If the solution in question is less dense to another solution, it is said to be hypobaric. If the two solutions have the same density, they are said to be isobaric with respect to one another. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
How will hypobaric solutions move when injected into the subarachnoid space?	Hypobaric solutions are less dense (lighter) than CSF and will 'float' to the highest possible anatomic position within the subarachnoid space. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
How will hyperbaric solutions move when injected into the subarachnoid space?	Hyperbaric solutions are more dense (heavier) than CSF and will 'sink' to the lowest possible position in the subarachnoid space. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
How will isobaric solutions move when injected into the subarachnoid space?	Isobaric solutions have a density that is equal to CSF and will generally remain in the position they are injected. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
What is the traditional technique to make a spinal anesthetic hypobaric?	Dissolving the drug in sterile water will result in a solution that is hypobaric in relationship to the CSF. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
What is the traditional technique for making a spinal anesthetic hyperbaric?	Dissolving the drug in 5% or 8% dextrose will make the solution hyperbaric. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*

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What is the traditional technique for making a solution isobaric?	Dissolving the solution in cerebrospinal fluid will result in an isobaric solution. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
If a hyperbaric spinal anesthetic is injected below the level of L3 and the patient is left in the sitting position, how will the local anesthetic migrate in the subarachnoid space?	The local anesthetic will drift inferiorly and result in a 'saddle block'. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1082.*
Theoretically, if a hyperbaric spinal anesthetic is injected at the L3 interspace and then the patient is positioned supine, how will the drug spread through the subarachnoid space and why?	Because the lumbar spinal curve peaks anteriorly at about L3, placing the patient supine would result in the spread of the drug in both a cephalad and caudal direction. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1082.*
What is the mechanism of action by which the addition of epinephrine to a spinal anesthetic prolongs its duration?	It is thought that the duration of action is prolonged by vasoconstriction and resultant decrease in the diffusion of the drug away from the site of action, by a direct nociceptive effect, or by a combination of these two actions. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
What is the total volume of cerebrospinal fluid at any given time in a normal adult?	About 150 mL. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 579.*
What is the normal CSF pressure?	CSF pressure is normally maintained between 10 and 20 cm H2O. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
About how much CSF is contained within the subarachnoid space?	Between 30 and 80 mL. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
How does the specific gravity of the cerebrospinal fluid change with age? Why does it change?	The specific gravity of the CSF tends to increase as the patient ages, primarily due to increases in glucose and protein content. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
What patient conditions may increase the specific gravity of the CSF? What patient conditions may decrease the specific gravity of the CSF?	The specific gravity of the CSF can increase with hyperglycemia and uremia and may decrease with liver disease. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*

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About how much cerebrospinal fluid is produced each hour?	About 30 mL per hour *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 690.*
About how much cerebrospinal fluid is produced each day?	500 mL *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
What factors are associated with an increased height of spinal block?	Conditions that result in increased abdominal pressure or engorgement of epidural veins such as pregnancy, ascites, abdominal tumors, kyphoscoliosis, or increased age reduce the CSF volume and increase the height of spinal blockade. In pregnant patients, the dose of spinal anesthetic to reach acceptable levels is one-third that of non-pregnant patients. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 956.*
How does the addition of sodium bicarbonate affect a spinal anesthetic?	Sodium bicarbonate 0.2 mL of a 0.42% solution is thought to shorten the onset of a spinal anesthetic, but does not prolong the duration of block. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 133.*
What doses of epinephrine, clonidine, or phenylephrine would be appropriate to mix with a spinal anesthetic to prolong the duration of the block?	Epinephrine 0.2 to 0.3 mg, clonidine 75 to 100 mcg, or phenylephrine 2 to 5 mg can be added to a spinal anesthetic to prolong the duration without resulting in any significant cardiovascular changes. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081-1082.*
How does the Taylor approach differ from the paramedian approach?	The Taylor approach is a lateral approach, but utilizes the L5-S1 interspace which is typically the largest interspace. The posterior superior iliac spine is identified and skin wheal is raised 1 cm medial and caudad to the inferior border of the posterior superior iliac spine. After local infiltration, the needle is directed 45 degrees medial toward the L5-S1 interspace. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 792.*
If you encounter bone during a subarachnoid block via the paramedian approach, how should you proceed?	You should walk the needle more cephalad and then medial until the subarachnoid space is encountered. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 792.*
If you encounter bone during a paramedian approach to a subarachnoid block, what specific bone have you most likely encountered?	The most common bone to encounter when performing a subarachnoid block via a paramedian approach is the caudad lamina. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 792.*

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How does a spinal anesthetic affect vital capacity? Tidal volume?	Vital capacity can diminish due to decreased intercostal muscle activity. The tidal volume is left unchanged. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 946.*
Why is urinary retention commonly associated with a subarachnoid block? Is urinary incontinence a complication of subarachnoid block?	The blockade of the autonomic fibers to the bladder from the S2-S4 nerve roots results in a decreased ability to void, so urinary retention is a common complication but urinary incontinence is not. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 947.*
How does a spinal anesthetic affect adrenal function?	Sympathectomy to the adrenal glands results in a decreased stress response to surgical stimulation which, by limiting the release of stress hormones, also limits any resulting increase in blood sugar. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 947.*
How does a spinal anesthetic affect the gastrointestinal system?	The sympathectomy produced by spinal anesthesia results in unopposed vagal activity. This increased parasympathetic activity results in increased peristalsis and a contracted gut. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 946-947.*
At what spinal levels would you expect a patient to complain of dyspnea?	A sensory level of T2-T4 can cause loss of sensory perception of abdominal wall and intercostal muscle movement resulting in a sensation of dyspnea despite normal respiration. A level of C2-C3 can result in phrenic nerve and intercostal muscle paralysis resulting in hypoxia, hypercarbia, or respiratory arrest. *Longnecker DE, Newman MF, Brown DL, Zapol WM. Anesthesiology. 2nd ed. New York: McGraw-Hill; 2012: 800.*
A patient with a T6 sensory level exhibits bradycardia and hypotension. What is the most likely cause?	In addition to the hypotension attributed to the increase in venous capacitance, blockade of the cardioacceleratory fibers at the T1 to T4 level can result in bradycardia and decreased myocardial contractility. Since a sympathetic blockade is typically 2 segments above a sensory block, a sensory block at T6 or higher could result in blockade of the cardioacceleratory fibers. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 945-946.*

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What are the causes of most allergic reactions to spinal anesthetics?	Most allergies are actually due to anxiety, epinephrine response, vasovagal reactions, or systemic toxic reactions to local anesthetics. The actual incidence of an anaphylactic reaction to a spinal anesthetic is 1 in 13,000. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 170.*
If a patient has a true allergic reaction to a spinal anesthetic, what are the most common presenting symptoms?	It is typically limited to a hypersensitivity reaction that results in erythema and/or edema of the mucous membranes. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 170.*
What local anesthetics are most predisposed to causing an allergic reaction and why?	The ester local anesthetics such as procaine, chloroprocaine, tetracaine, and benzocaine are most commonly implicated in IgE-mediated allergic reactions. The higher risk is due to the para-aminobenzoic acid preservative to which patients have had previous exposure. PABA is found in common items such as lotions, sunscreens, cosmetics, sulfonamides, and some foods. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 170.*
What is the incidence and cause of cardiac arrest during spinal anesthesia?	The incidence of cardiac arrest in spinal anesthesia is about 1 in 4,000. The cause is blockade of the cardioacceleratory fibers from T1 to T5 and a drop in venous return. It is usually preceded by bradycardia and hypotension. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 171.*
What is transient neurologic syndrome?	TNS, which has historically been referred to as transient radicular irritation, is defined as pain in the lower back or buttocks that may radiate to one or both legs after a spinal anesthetic. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 172.*
How long does transient neurologic syndrome typically last?	About 1 week. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 172.*
What local anesthetics are associated with the highest incidence of transient neurologic syndrome?	Mepivicaine and lidocaine have the highest association with TNS and bupivacaine has the lowest incidence. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 173.*

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What is cauda equina syndrome?	Cauda equina syndrome consists of lower back pain, sciatica, motor and sensory loss, and bladder and bowel dysfunction that typically occurs due to trauma, lumbar disc disease, ankylosing spondylitis, tumors, or abscesses in the lumbar area. It has also been associated with prolonged exposure of the cauda equina to high doses or high concentrations of local anesthetics that cause direct neurotoxicity. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 173.*
What is arachnoiditis and what causes it?	Arachnoiditis in an inflammatory disorder of the arachnoid mater which surrounds the spinal cord and cauda equina. It can be caused by exposure of the arachnoid membrane to povidone-iodine solution, vasoconstrictors, local anesthetics, blood, and contrast media. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 173.*
Why is the cauda equina susceptible to injury?	The cauda equina is composed of nerve roots below the terminal level of the spinal cord. Because these nerves have an underdeveloped epineurium and are hypovascular relative to the rest of the cord, they are more susceptible to injury. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 173.*
What are the symptoms of arachnoiditis?	Lower back and leg pain that increases with activity that may include a wide range of sensory and motor alterations. It usually is not progressive, but significant improvement is not likely. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 173.*
What is anterior spinal artery syndrome?	Anterior spinal artery syndrome is characterized by painless loss of motor and sensory function due to interruption of the blood supply through the anterior spinal artery. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 178.*
How is anterior spinal artery syndrome linked to spinal anesthesia?	Prolonged hypotension due to a spinal anesthetic has been linked as a causative factor in the development of ASAS. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 178.*
Why does anterior spinal artery syndrome not affect the sense of vibration and proprioception in the affected areas?	The anterior spinal artery supplies the anterior two-thirds of the spinal cord. The nerves responsible for proprioception and vibration sense are located in the posterior spinal cord. *Wong, CA. Spinal and Epidural Anesthesia. New York: McGraw-Hill; 2007: 178.*

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How does total spinal anesthesia occur and what are its symptoms?	Total spinal anesthesia occurs when the local anesthetic spreads throughout the subarachnoid space high enough to block sympathetic outflow from the entire spinal cord and possibly even the brainstem. It results in respiratory arrest, profound hypotension, and bradycardia from sympathetic blockade. Spinal blockade, even at the cervical level is reported not to be high enough to block the large A-alpha fibers of the phrenic nerve. Apnea appears to be the result of decreased perfusion to the brainstem as apnea typically resolves with hemodynamic resuscitation. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 927.*
How is total spinal blockade treated?	Respiratory support, vasopressors, IV fluids, and atropine are administered as needed. If cardiopulmonary status is maintained, a total spinal block is self-resolving as the block recedes. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 927-928.*
What factors increase the risk for postdural puncture headache (PDPH) when performing spinal anesthetics?	The incidence of postdural puncture headache increases with the use of non-pencil point needles, a cutting needle inserted with the bevel of the needle perpendicular to the long axis of the body, and multiple attempts. It is also more likely to occur in young patients than elderly patients and more common in female patients. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1085.*
How much epinephrine would you add to a spinal anesthetic dose to prolong the duration of the block?	0.1 to 0.2 mL of 1:1000 (1 mg/mL) epinephrine is the dose typically added to the local anesthetic to prolong the duration of the spinal. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
How does the addition of epinephrine comparatively affect the duration of action of tetracaine, lidocaine, and bupivacaine?	The addition of epinephrine prolongs the duration of tetracaine the most, lidocaine moderately, and has almost no effect on the duration of action of bupivacaine. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1081.*
What opioids are typically added to subarachnoid block injections and what doses are used?	Fentanyl 10-25 mcg, sufentanil 2.5-10 mcg, preservative-free morphine 250 mcg, or even clonidine 150 mcg, all of which act to prolong the duration of the spinal anesthetic. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1082.*

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Subarachnoid Block

What is a saddle block?	A saddle block is a hyperbaric spinal anesthetic in which the patient remains sitting after injection for 5 minutes so that the local anesthetic can concentrate on the lumbar and sacral nerves. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1082.*
In what manner does a spinal block wear off?	A spinal block does not end abruptly, nor does sensation throughout the affected dermatomes return at the same rate. A spinal anesthetic recedes gradually from superior to inferior. As a result, the sacral dermatomes demonstrate a longer duration of action than do the lumbar or thoracic dermatomes. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 919.*
From shortest to longest, rank procaine, tetracaine, bupivacaine, lidocaine, and chloroprocaine in order of the duration of their spinal anesthetic when equivalent dosages are administered.	Chloroprocaine and procaine are the shortest acting agents with durations around 90-120 minutes. Lidocaine has a duration around 140-240 minutes. Bupivacaine and tetracaine exhibit durations around 240-380 minutes. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 919.*
A patient with a T8 sensory level following a spinal anesthetic exhibits hypotension. Is this due to dilation of venous or arterial vessels?	The sympathectomy associated with a spinal anesthetic produces hypotension due principally to venous pooling from dilation of the venous capacitance vessels, although arteriolar dilation may contribute to the hypotension by a lesser degree. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 946.*

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